

PrimeSim™ Help: What's New

Version W-2024.09, September 2024



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What's New in This Release

PrimeSim version W-2024.09 includes the following new features, enhancements, and changes.

- [Enhancement: Additional New or Updated Commands and Options](#)
- [Additional Product and Documentation Enhancements](#)
- [GPU Hardware and Software Requirements Updates](#)

Enhancement: Additional New or Updated Commands and Options

Additional new or updated PrimeSim netlist options and commands:

- Added `primesim_ckpt_report_limit` option - Controls the maximum number of error reports generated for each file.chkdevop file during simulation.
- Added `primesim_output_op` option - Sets the output format for the operating point (.op) analysis.
- Added `primesim_spf_include` option - Includes only the specified instances from SPF back-annotation and while ignoring the rest of the instances
- Added `primesim_analog` option - Provides a turnkey accuracy setting to run large post-layout analog centric circuits in PrimeSim Pro.
- Added multi-CPU command-line option `-dpincremental` - Reruns a Monte Carlo simulation for unfinished or new samples. After rerun, it automatically combines the measure results from the original runs.
- PrimeSim command line argument `-dscale` is enhanced to support a new usage format as:

```
primesim -dscale output.dynamic -dp number
```

Additional new or updated PrimeSim SPICE netlist commands and control options:

- Updated the `.NOISE` command with new argument `SEPARATE_NOISE` that specifies whether to separate output noise contribution from any noise source into input noise and noise gain.

For more information, see the PrimeSim™ User Guide: Pro and SPICE.

Additional Product and Documentation Enhancements

- Enhanced PrimeSim to support `PSFXL` output format for transient analysis.
- Added support for `zstd` compression file format.
- PrimeSim SPICE out-of-the-box (OOTB) enhancements on:
 - Transient simulation performance
 - S-parameter simulation robustness, performance, and memory usage
 - DCOP robustness, ease of use convergence
 - Small-signal analysis robustness and performance
- Updated “Distributed Processing Configuration Files” table in “Running Distributed Processing on a Network Grid” topic to add information related to “CUSTOM” protocol.
- Updated “Ease-of-Use Settings for runlvl” and “PrimeSim runlvl for analog circuit simulation” tables in “PrimeSim SPICE Accuracy and Speed Options (-runlvl and .option runlvl)” topic.
- Added `primesim_case_sync_in_subckt` option information in the user guide.
- Added `-spectre` command-line option information in the user guide.

For more information, see the PrimeSim™ User Guide: Pro and SPICE.

GPU Hardware and Software Requirements Updates

In version W-2024.09, recommended GPU driver and OS have been updated:

- GPU Driver: 535.183.01 and newer versions
- OS: AlmaLinux 8.6 and newer versions

For more information, see the GPU Hardware and Software Requirements in PrimeSim™ User Guide: Pro and SPICE.